PHYSIOLOGY OF SMOOTH MUSCLE

Smooth Muscles

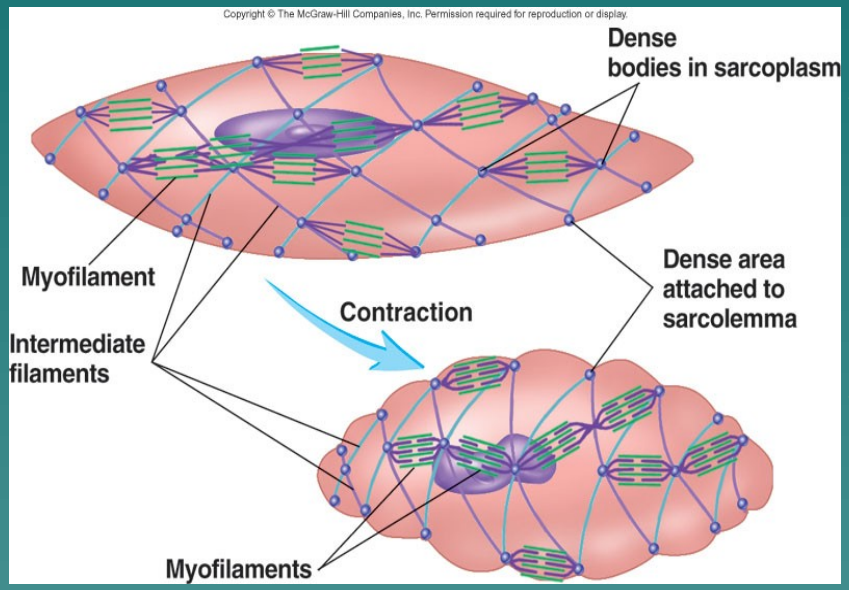
LEARNING OBJECTIVES

- Structure & Properties of Smooth Muscles
- Describe mechanism of contraction
- Molecular basis of contraction of smooth muscle
- Differentiate smooth muscle with cardiac and skeletal muscle

Smooth Muscles

- Location: In the walls of hollow organs, blood vessels, eye, glands, uterus, skin
- Some functions: propel urine, mix food in digestive tract, dilating/constricting pupils, regulating blood flow
- In some locations, autorhythmic
- Controlled involuntarily by endocrine and autonomic nervous systems

Smooth Muscles



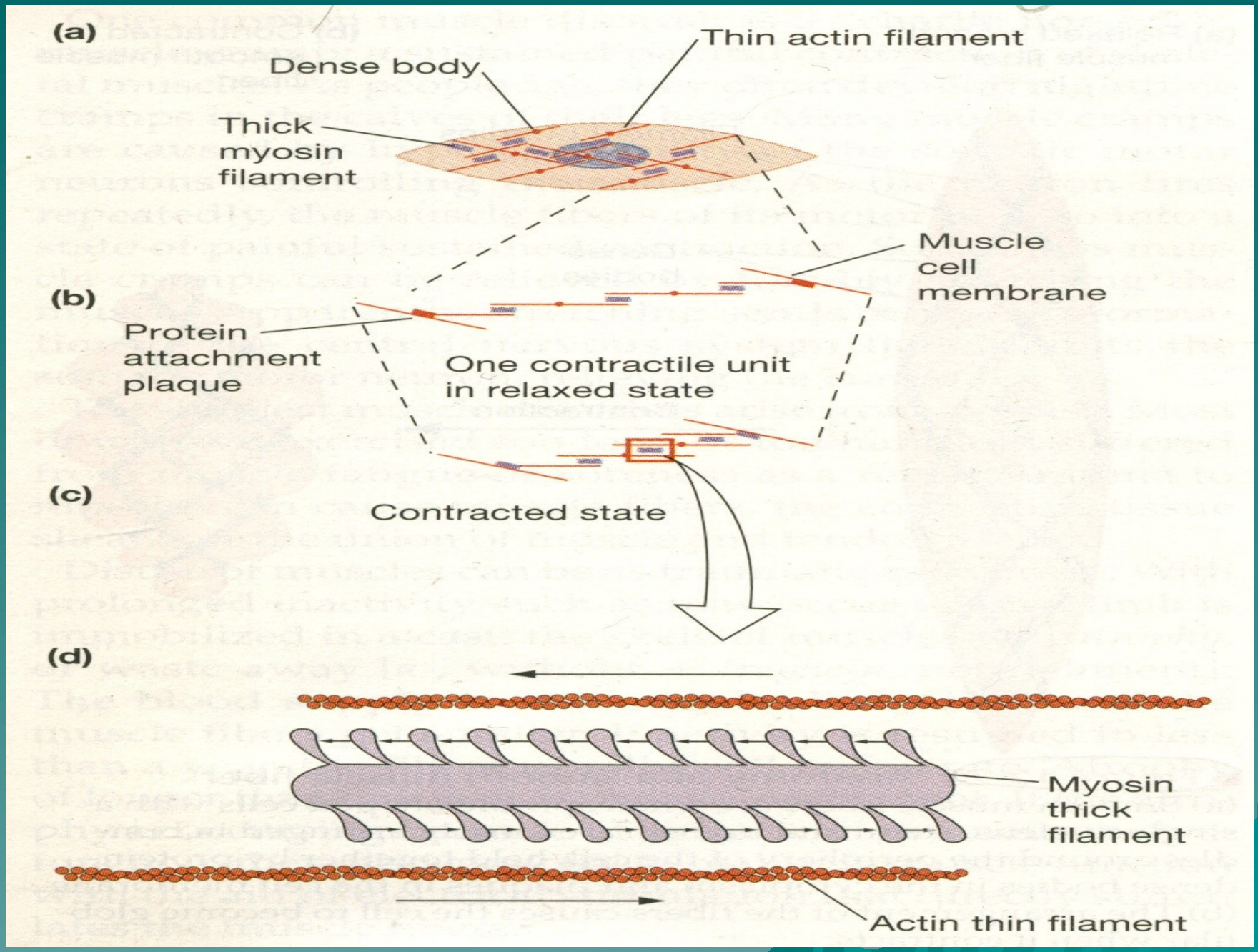
- Fibers smaller than those in skeletal muscle
- Involuntary
- Slow, wave-like contractions
- Spindle shaped
  - 1-5 µm in diameter
  - 20-500 µm in length
- A single **nucleus** present in the central thick portion.

Smooth Muscles

- Cytoplasm appears homogenous without striations.
- Fewer mitochondria as compared to the skeletal muscle.

- **Dense bodies** present (instead of Z disks) attached to the cell membranes OR dispersed throughout the cell

- Attached to the dense bodies are numerous numbers of **Actin filaments**


(a)
Thin actin filament
Dense body
Thick
myosin
filament
Muscle
cell
membrane
(b)
Protein
attachment
plaque
One contractile unit
in relaxed state
(c)
Contracted state
(d)
Myosin
thick
filament
Actin thin filament

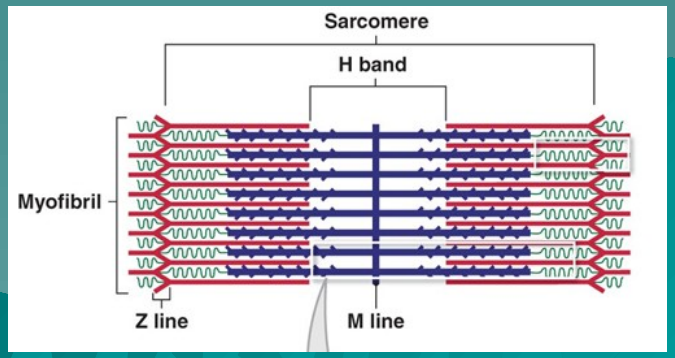
Smooth Muscles

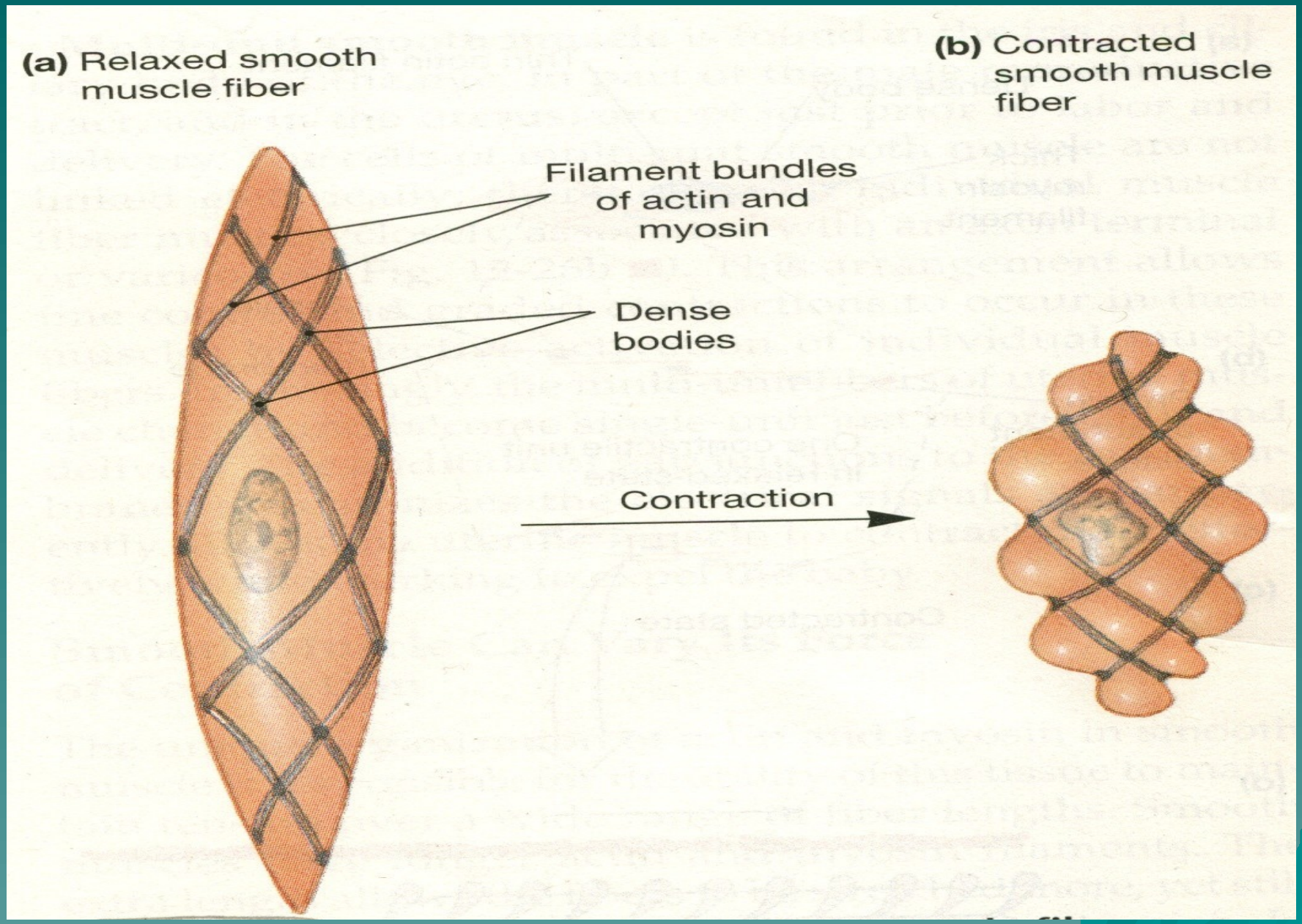
- Interspersed between the actin filaments are **Myosin filaments** (their diameter twice as much as actin filaments)

  →*Usually, 5-10 times as many actin filaments as Myosin filaments*

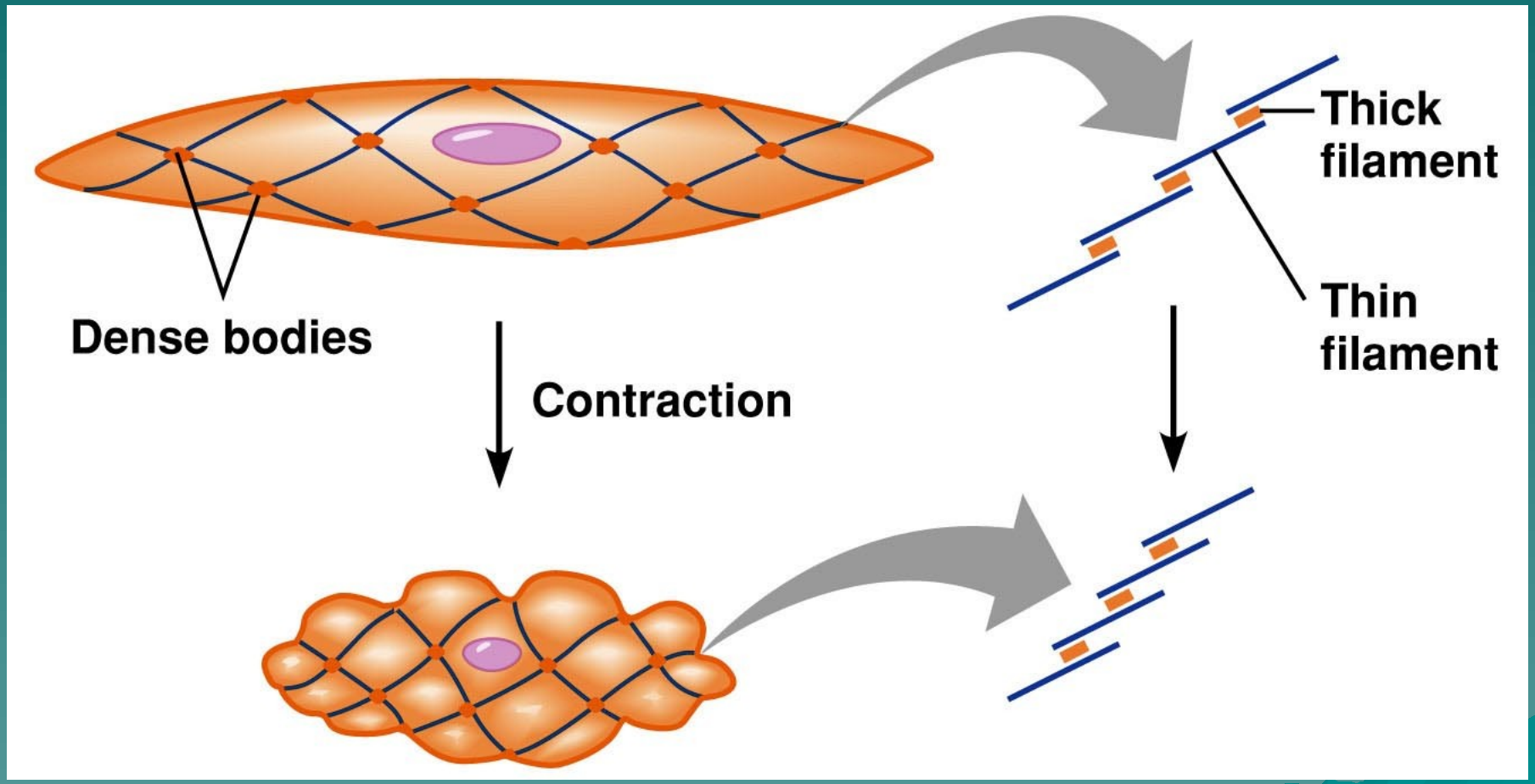
- No sarcomeres
  - Not arranged as symmetrically as in skeletal muscle, thus NO striations.

Skeletal muscle




(a) Relaxed smooth muscle fiber
(b) Contracted smooth muscle fiber
Filament bundles of actin and myosin
Dense bodies
Contraction

Smooth Muscle Cell



Smooth Muscles

Two types of Smooth muscles

1-Multi unit smooth muscle

2-Unitary smooth muscle

(SINGLE UNIT or SYNCYTIAL or VISCERAL Smooth muscle)

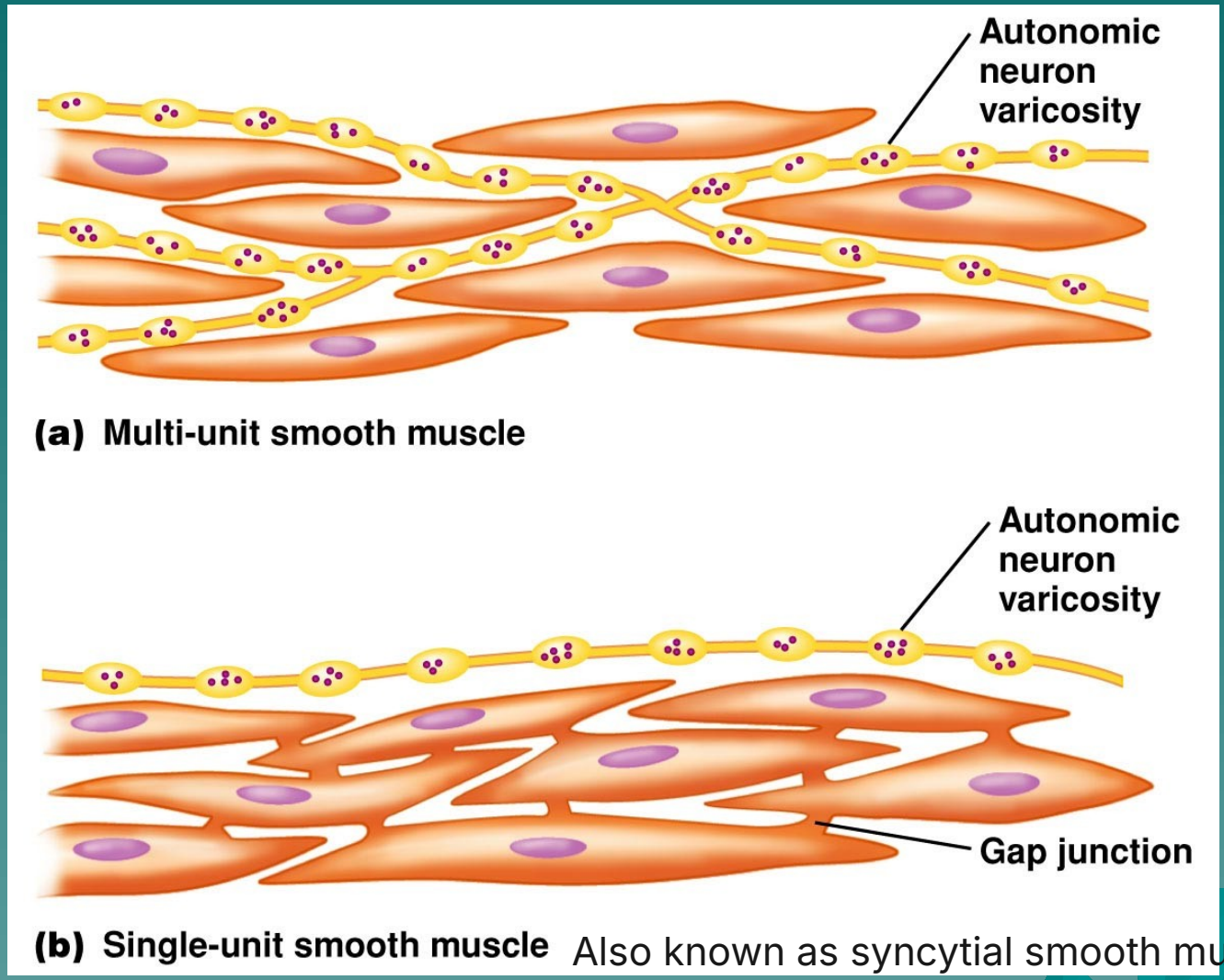
Multi unit smooth muscle

- Composed of discrete, separate smooth muscle fibers
- Each fiber operates independently of the others and often is innervated by a single nerve ending
- Each fiber can contract independently of the others – MOST IMPORTANT PROPERTY
- Control is exerted mainly by nerve signals

Unitary smooth muscle

- Composed of a mass of hundreds to thousands of smooth muscle fibers that contract together as a single unit
- The fibers usually are arranged in sheets or bundles
- The cell membranes are joined by many *gap junctions*

Multi *vs*. Single-Unit Muscle



Also known as syncytial smooth muscle

CLASSIFICATION OF SMOOTH MUSCLES

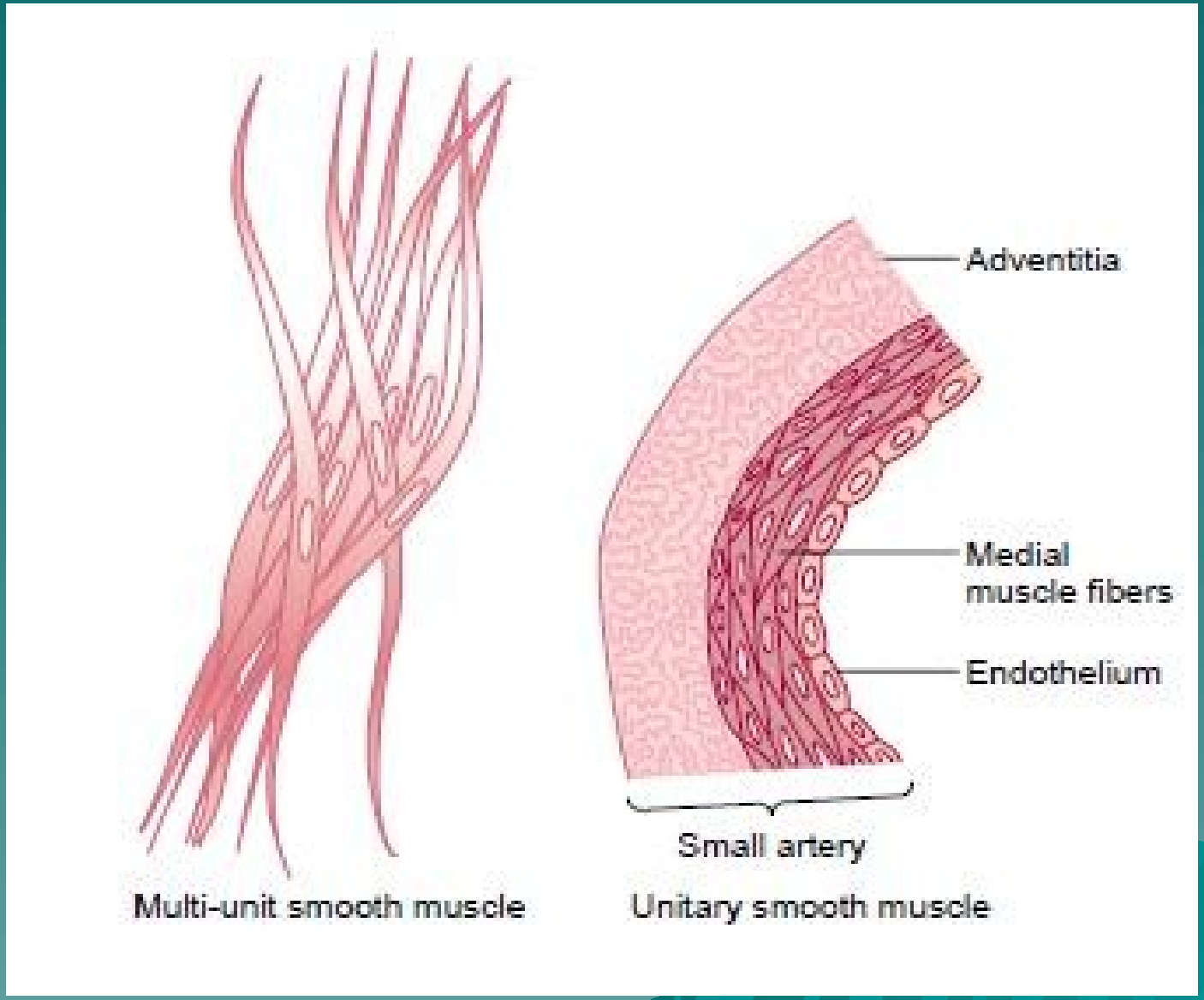
UNITARY/ SINGLE UNIT/SYNCYTIAL/VISCERAL

1. Muscles of visceral organs .e.g. GIT, uterus, ureters & some of the smaller blood vessels.
2. Form a sheet or bundles of tissue.
3. Cell membranes show gap junctions that allows AP to pass rapidly from cell to cell.
4. AP spreads rapidly throughout the sheet of cells – ***cells contract as a single unit.***

MULTI-UNIT

1. Iris & Ciliary body of the eye, large arteries, Piloerector muscles
2. Showing discrete, individual smooth muscle fibers.
3. Smooth muscle cells not electrically linked. Each muscle fiber innervated by a single nerve ending. NT itself can spread and lead to an AP.
4. Selective activation of each muscle fiber that can then contract independently of each other.

Multi unit and unitary smooth muscles



Smooth muscle contraction

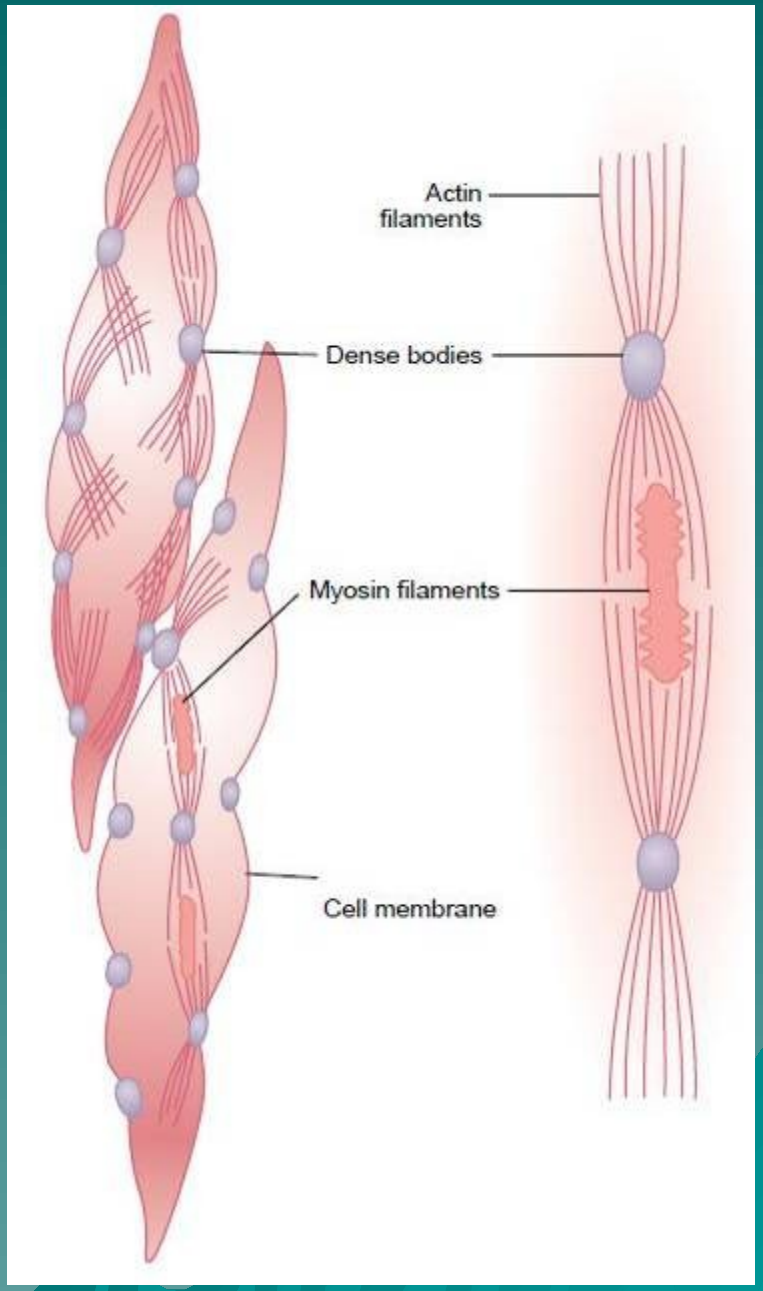
- *Actin* and *myosin filaments*, similar to those found in skeletal muscle.
- Does not contain the normal troponin complex
- Actin and myosin filaments interact with each other in much the same way that they do in skeletal muscle.
- Contractile process is activated by calcium ions, and ATP is degraded to ADP to provide the energy for contraction.

- Smooth muscle does not have the same striated arrangement of actin and myosin filaments as is found in skeletal muscle
- Actin filaments attached to so-called ***dense bodies*** *which* are attached to the cell membrane.Others are dispersed inside the cell.
- Interspersed among the actin filaments in the muscle fiber are myosin filaments.

Actin filaments radiate away from the dense bodies and between the two dense bodies ,are found the myosin filaments , thus forming a contractile unit of smooth muscle.

This contractile unit is similar to the contractile unit of skeletal muscle, but without the regularity of the skeletal muscle structure

The dense bodies of smooth muscle serve the same role as the Z discs in skeletal muscle.



STRUCTURE OF SMOOTH MUSCLE

SIDEPOLAR CROSS-BRIDGES:

Myosin filaments have sidepolar cross-bridges

↓

Bridges on one side hinge in one direction & on the other side in the opposite direction

↓

Allows myosin to pull an actin filament in one direction while simultaneously pulling it in the other direction on the other side

↓

Allow smooth muscle to contract 80% as compared to only 30 % in the skeletal muscle

(force of contraction in skeletal muscle is limited because of the presence of the z-disc, against which the thick filament will abut against and cannot move any further)

- Calcium Pump: pumps Ca back into the SR if present for relaxation to take place. But it is very slow so that duration of cont. is longer.

Whats new in smooth muscle contraction ?

- The rapidity of cycling of the myosin cross-bridges in smooth muscle—their attachment to actin, then release from the actin, and reattachment for the next cycle is much slower in smooth muscle
- The *fraction of time* that the cross-bridges remain attached to the actin filaments, which determines the force of contraction, is greatly increased in smooth muscle
- Only 1/10 to 1/300 as much energy is required to sustain the same tension of contraction in smooth muscle as in skeletal muscle.

“Latch” Mechanism

- It is a state in which the dephosphrylated myosin remains attached to actin for prolonged period of time. This produces sustained contraction without consuming ATP & thus enables the smooth muscle to sustain long-term maintenance of tone without fatigue. E.g. urinary bladder full of urine.

- Further, the energy consumed to maintain contraction is as little as 1/300 the energy required for comparable sustained skeletal muscle contraction. This is called the “latch” mechanism.

- **The importance of the latch mechanism** is that it can maintain prolonged tonic contraction in smooth muscle for hours with little use of energy

Stress-Relaxation of Smooth Muscle.

- Definition:

  Ability to return to nearly its original *force* of contraction seconds or minutes after it has been elongated or shortened.

This phenomenon is found especially in the visceral unitary type of smooth muscle of many hollow organs

Regulation of Contraction by Calcium Ions

- Contraction in Smooth muscle is initiated by the rising conc. Of intracellular calcium
- This increase can be caused in different types of smooth muscle by
  - -nerve stimulation
  - -hormonal stimulation
  - -stretch of the fiber
  - -change in the chemical environment of the fiber.

Increased Influx of Ca into the cell can be from the following sources

1. ECF thru Ca channels
2. Ca released from SR
3. Stretch-activated Ca channels when memb. Deformed

ROLE OF CALCIUM

- Poorly developed SR
- Presence of **caveolae**- Small invaginations abut the SR which release Ca when AP reaches it.

Thus, smooth muscle contraction is highly dependent on **Extracellular Calcium conc.**

Point to Note:

So the main source of Calcium ions in smooth muscle is to greater extent ECF and to a lesser extent SR as compared to the skeletal muscles where greatest source of Calcium is SR.

Calcium plays the main role in the prolonged contraction process.

SMOOTH MUSCLE CONTRACTION SEQUENCE OF EVENTS:

Binding of Ach to the receptors

↓

Increased Influx of Ca into the cell

↓

Ca binds to **Calmodulin**

↓

Ca-Calmodulin activates the enzyme: Myosin light chain kinase MLCK or simply **Myosin kinase**

↓

Phosphorylation of myosin, using energy & Pi from ATP

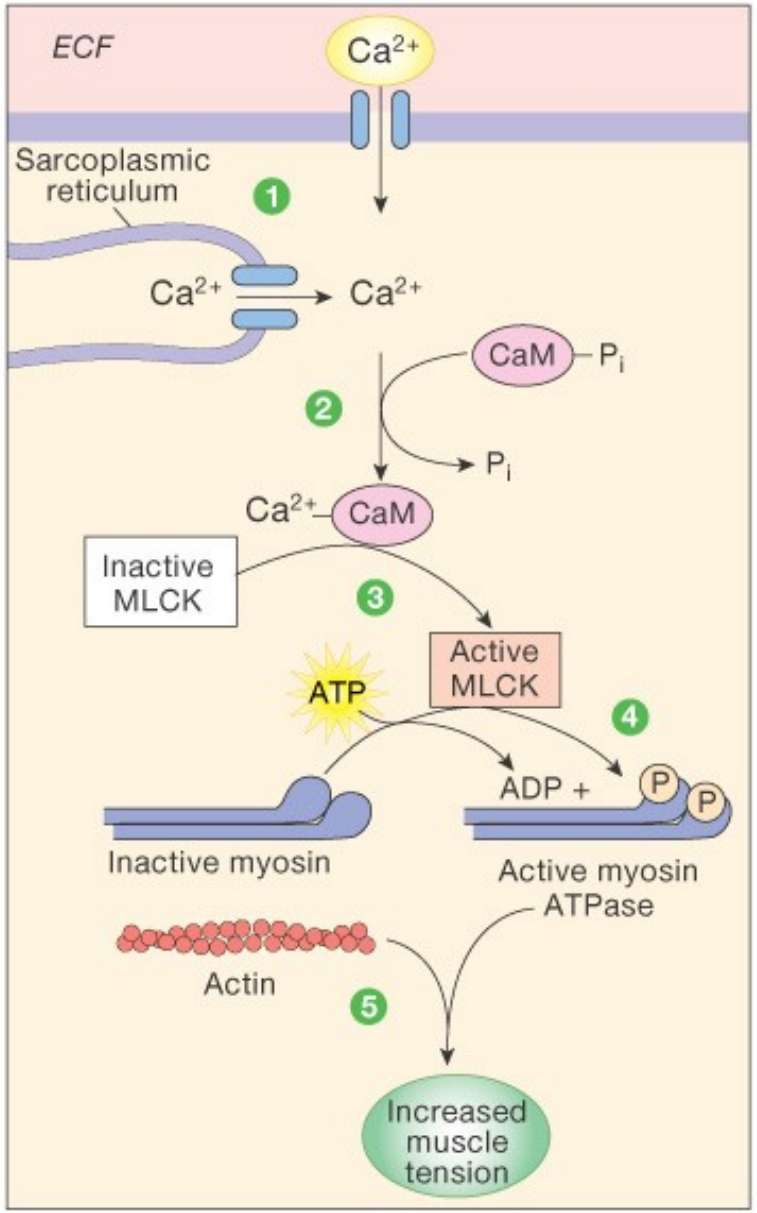
↓

Increased ATPase activity & binding of myosin to actin

↓

Contraction of smooth muscle

Smooth Muscle Contraction: Mechanism



1. Intracellular Ca^{2+} concentrations increase when Ca^{2+} enters cell and is released from sarcoplasmic reticulum.
2. Ca^{2+} binds to calmodulin (CaM).
3. Ca^{2+}–calmodulin activates myosin light chain kinase (MLCK).
4. MLCK phosphorylates light chains in myosin heads and increases myosin ATPase activity.
5. Active myosin crossbridges slide along actin and create muscle tension.

SMOOTH MUSCLE RELAXATION SEQUENCE OF EVENTS:

Dephosphorylation of Myosin by **myosin phosphatase**/ MLCP

↓

Decreases its ATP activity

↓

Ca removed from cytoplasm using Ca-Na antiport protein & Ca-ATPase

↓

Calmodulin releases Ca & uncomplexes from MK

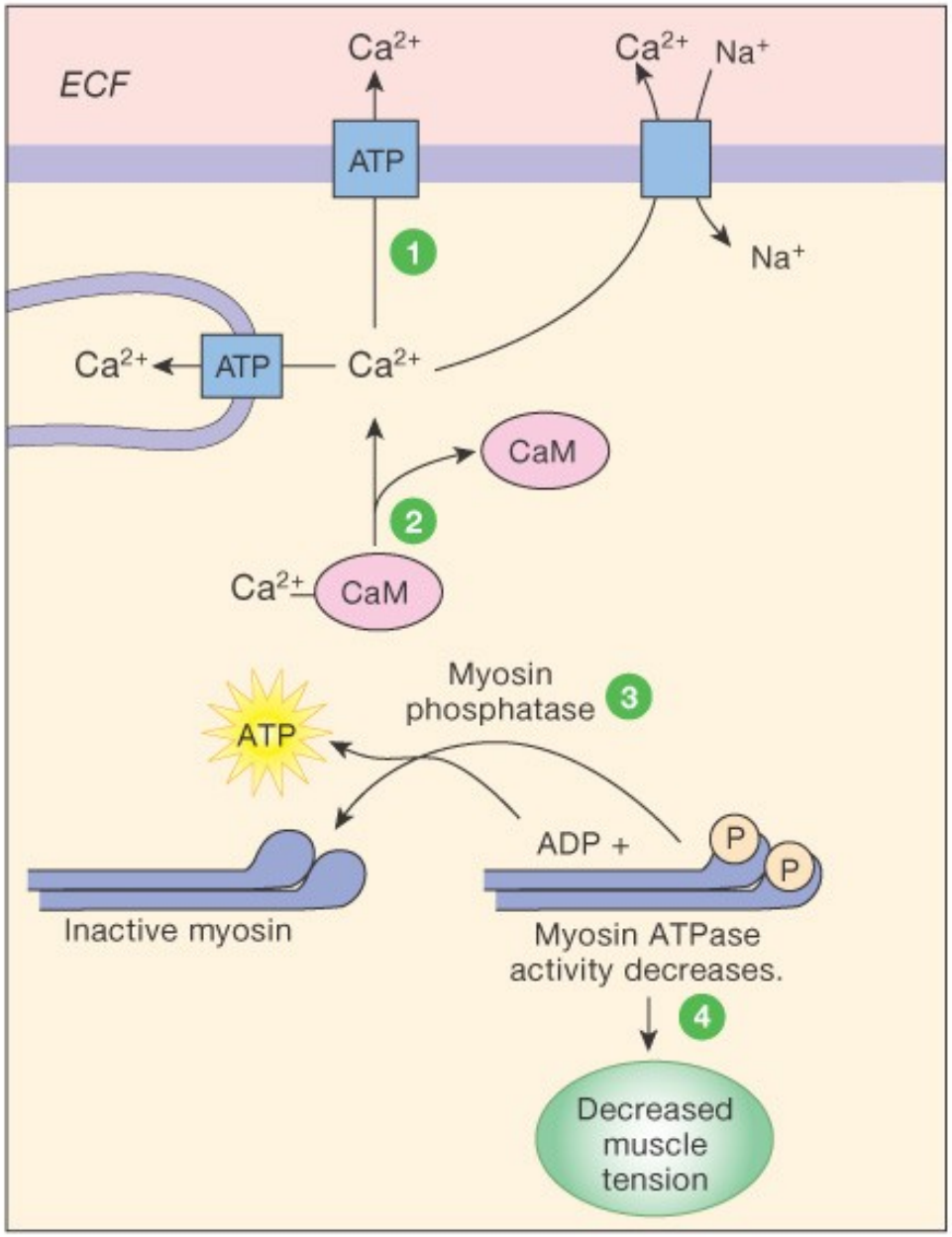
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MK is phosphorylated by **Protein kinase**, inactivating it

↓

Relaxation OR sustained contraction

Smooth Muscle Relaxation: Mechanism



1. Free Ca^{2+} in cytosol decreases when Ca^{2+} is pumped out of the cell or back into the sarcoplasmic reticulum.

2. Ca^{2+} unbinds from calmodulin (CaM).

3. Myosin phosphatase removes phosphate from myosin, which decreases myosin ATPase activity.

4. Less myosin ATPase results in decreased muscle tension.

Neuromuscular Junctions of Smooth Muscle

- **Autonomic nerve fibers** that innervate smooth muscle generally branch diffusely on top of a sheet of muscle fibers
- In most instances, these fibers do not make direct contact with the smooth muscle fiber cell membranes but instead form so-called **diffuse junctions** that secrete their transmitter substance into the matrix coating of the smooth muscle away from the muscle cells
- The transmitter substance then diffuses to the cells.

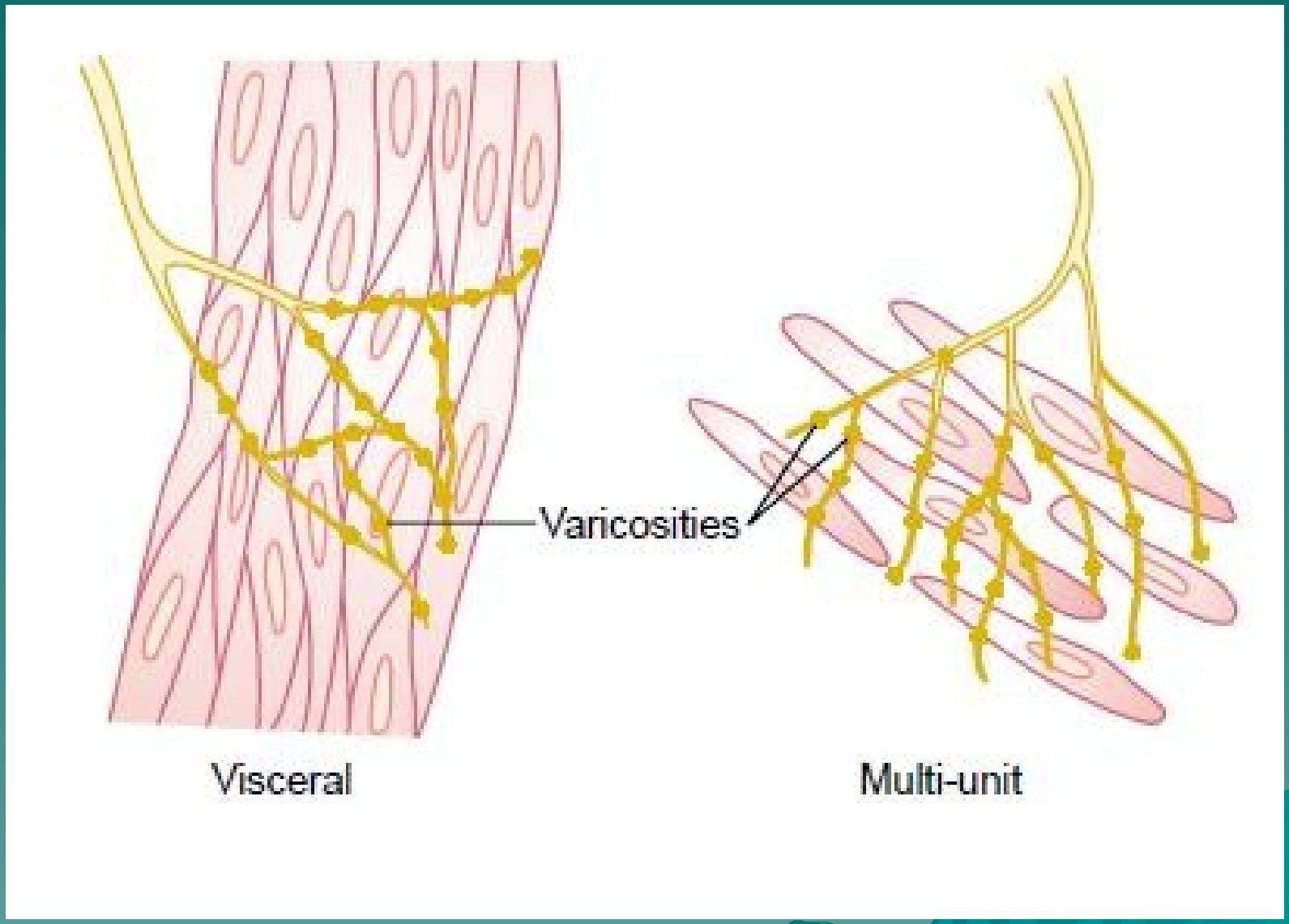
- The axons that innervate smooth muscle fibers do not have typical branching end feet of the type in the motor end plate on skeletal muscle fibers.
- Instead, most of the fine terminal axons have multiple *varicosities* along their axes.
- At these points the *Schwann cells* that envelop the axons are interrupted so that transmitter substance can be secreted through the walls of the varicosities
- The vesicles in the varicosities of the autonomic nerve fiber endings contain *acetylcholine* in some fibers and *norepinephrine* in others—and other substances as well.

- In the multi-unit type of smooth muscle, the varicosities are separated from muscle cell membrane by as little as 20 to 30 nanometers—the same width as the synaptic cleft that occurs in the skeletal muscle junction.

- These are called **contact junctions**, and they function in much the same way as the skeletal muscle neuromuscular junction

- The rapidity of contraction of these smooth muscle fibers is considerably faster than that of fibers stimulated by the diffuse junctions

Innervation of Smooth muscle



ACTION POTENTIAL

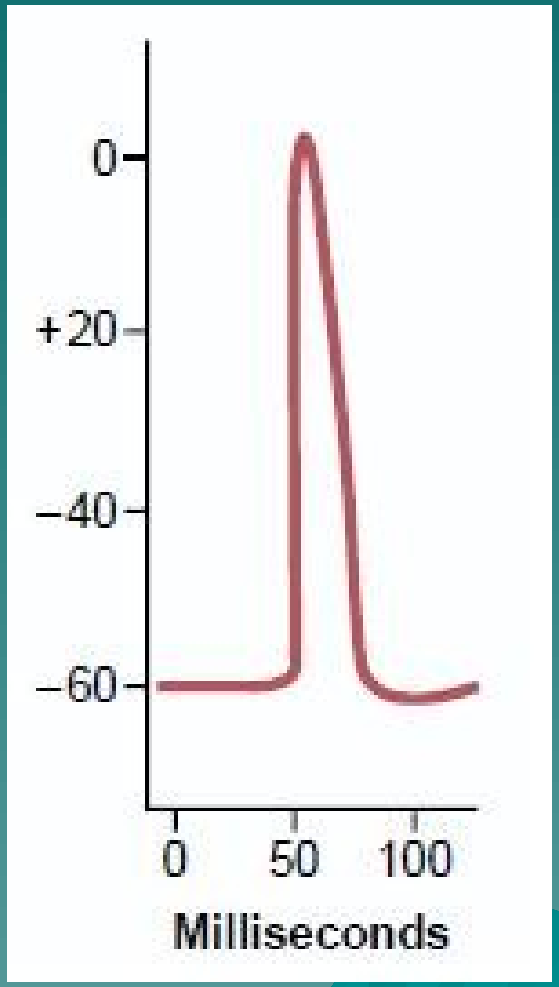
- In the normal resting state, the membrane potential is about -50 to -60 mv. which is about 30 millivolts less negative than in skeletal muscle.

- The AP of visceral or **Unitary** smooth muscle is of 2 types:

1. **Typical spike potentials**: (similar to skeletal muscles) -mostly seen in the unitary smooth muscles
2. **AP with Plateaus**: Starts like a typical spike potential but repolarization delayed for several hundred to as many as 1000 msec ----accounts for the prolonged contraction that occurs in certain organs

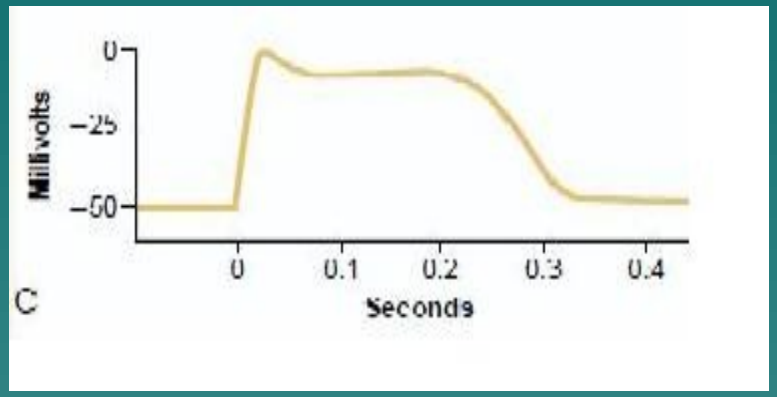
Spike Potentials.

- Spike action potentials, such as seen in skeletal muscle, occur in most types of unitary smooth muscle.
- The duration of this type of action potential is 10 to 50 milliseconds
- Such action potentials can be elicited in many ways, for example, by electrical stimulation, by the action of hormones on the smooth muscle, by the action of transmitter substances from nerve fibers, by stretch, or as a result of spontaneous generation in the muscle fiber itself.



Action Potentials with Plateaus.

- The onset of this action potential is similar to that of the typical spike potential.
- However, instead of rapid repolarization of the muscle fiber membrane, the repolarization is delayed for several hundred to as much as 1000 milliseconds (1 second).
- The importance of the plateau is that it can account for the prolonged contraction that occurs in some types of smooth muscle, such as the ureter, the uterus under some conditions, and certain types of vascular smooth muscle.



ACTION POTENTIAL

SLOW WAVE POTENTIALS

Without an external stimulus membrane potential is often associated with a basic slow wave rhythm. This is itself not an AP but a local property of the smooth muscle fibers.

CAUSE:

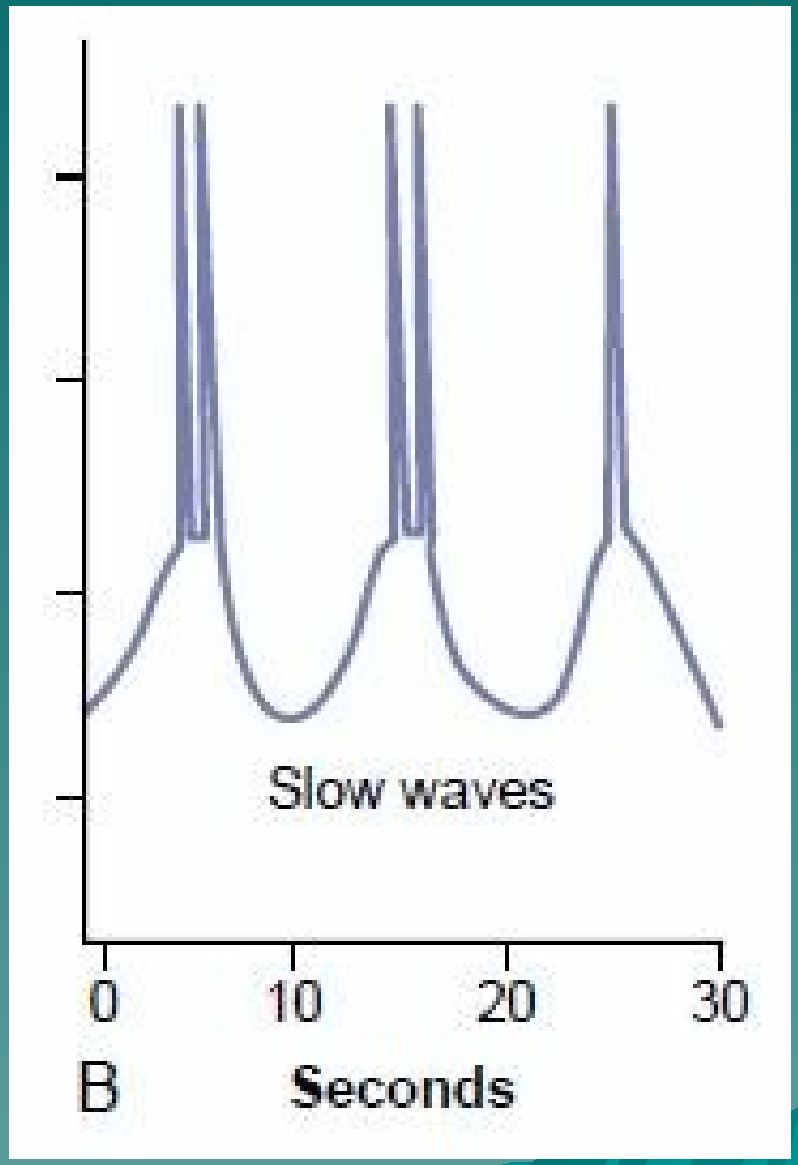
1. Waxing & waning of the pumping of Na ions
2. Conductance of the ion channels increase & decrease rhythmically

IMPORTANCE:

When the peak of the slow wave reaches about -35 mv, threshold is reached and an AP develops & leads to a contraction.

Thus, at peak of the slow waves an AP can occur. These slow waves are called as **Pacemaker waves.**

Slow waves



PROPERTIES OF CARDIAC MUSCLES

LECTURE OBJECTIVES

By the end of this lecture students should be able to:

- Define the type of cardiac muscle cells
- Define the different properties of cardiac muscles
- Understand the implications of the properties of cardiac muscles
- Understand the mechanism of pacemaker activity of the heart

TYPES OF CARDIAC MUSCLES

There are two types of cardiac muscles:

- Structural/ contractile
- Pace maker cells

CARDIAC MUSCLE CELLS

CONTRACTILE CELLS:

- Similar to skeletal muscle
- 3 differences:

1. Longer action potential
2. Entry of extracellular calcium ions through voltage-regulated calcium channels
   - *Delay depolarization*
   - *Initiate contraction*
   - *Trigger release of Ca2+ from reserves in sarcoplasmic*
   - *reticulum*
3. Longer contraction

PROPERTIES OF CARDIAC MUSCLE CELLS

- **Contractile cells**: Contractility
- **Excitable cells**: Automacity
- *Twitch summation does not occur: Tetanus is not possible*

Automaticity (Autorhythmicity)

The ability to produce spontaneous rhythmic excitation without external stimulus.

(1) Intrinsic rhythm of excitable cells

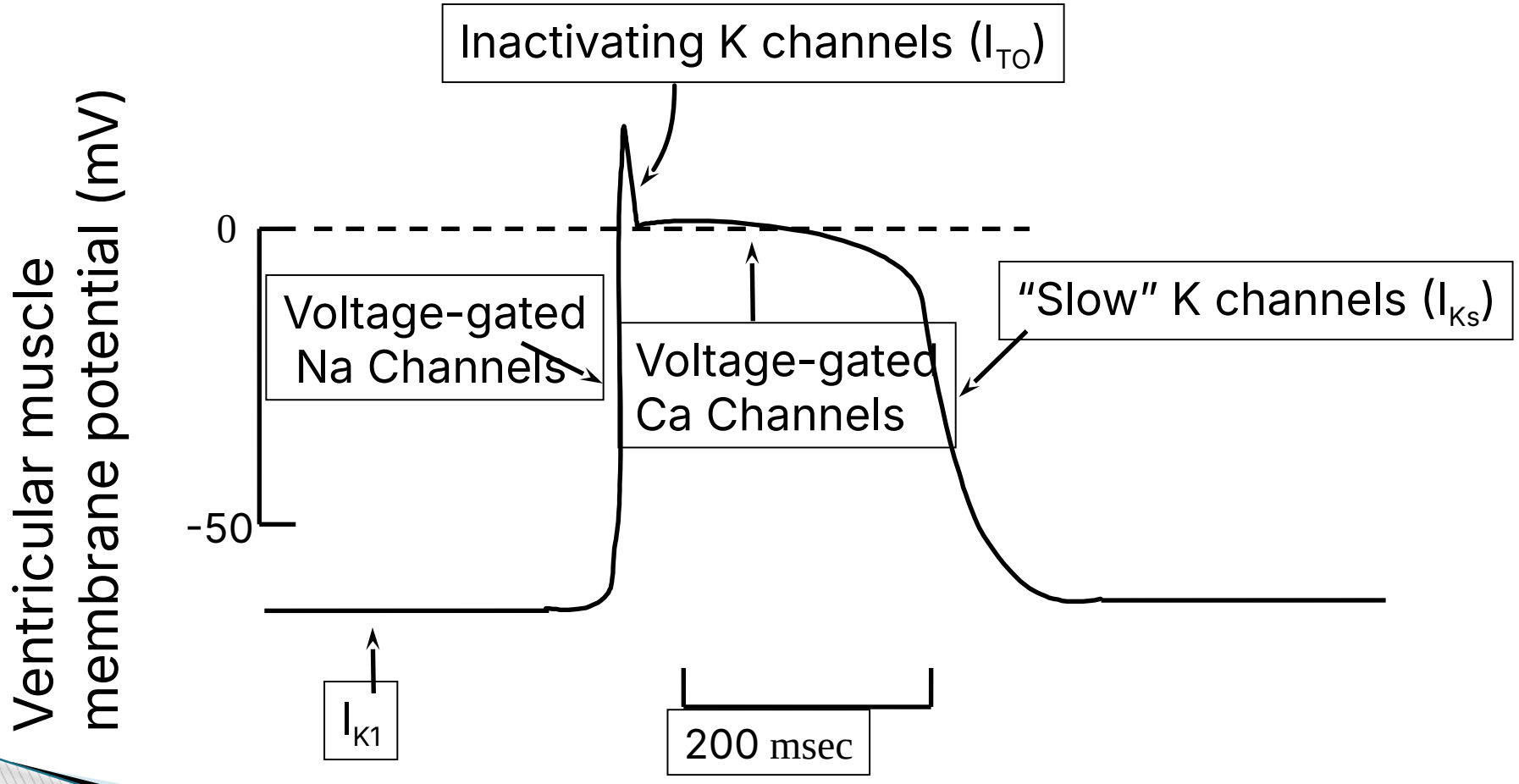
- Purkinje fiber: **15 – 40 /min**
- Atrioventricular node: **40 – 60 /min**
- Sinoatrial node: **90 – 100 /min**

(2) NORMAL PACEMAKER OF THE HEART: **SA NODE (SINO ATRIAL NODE)**

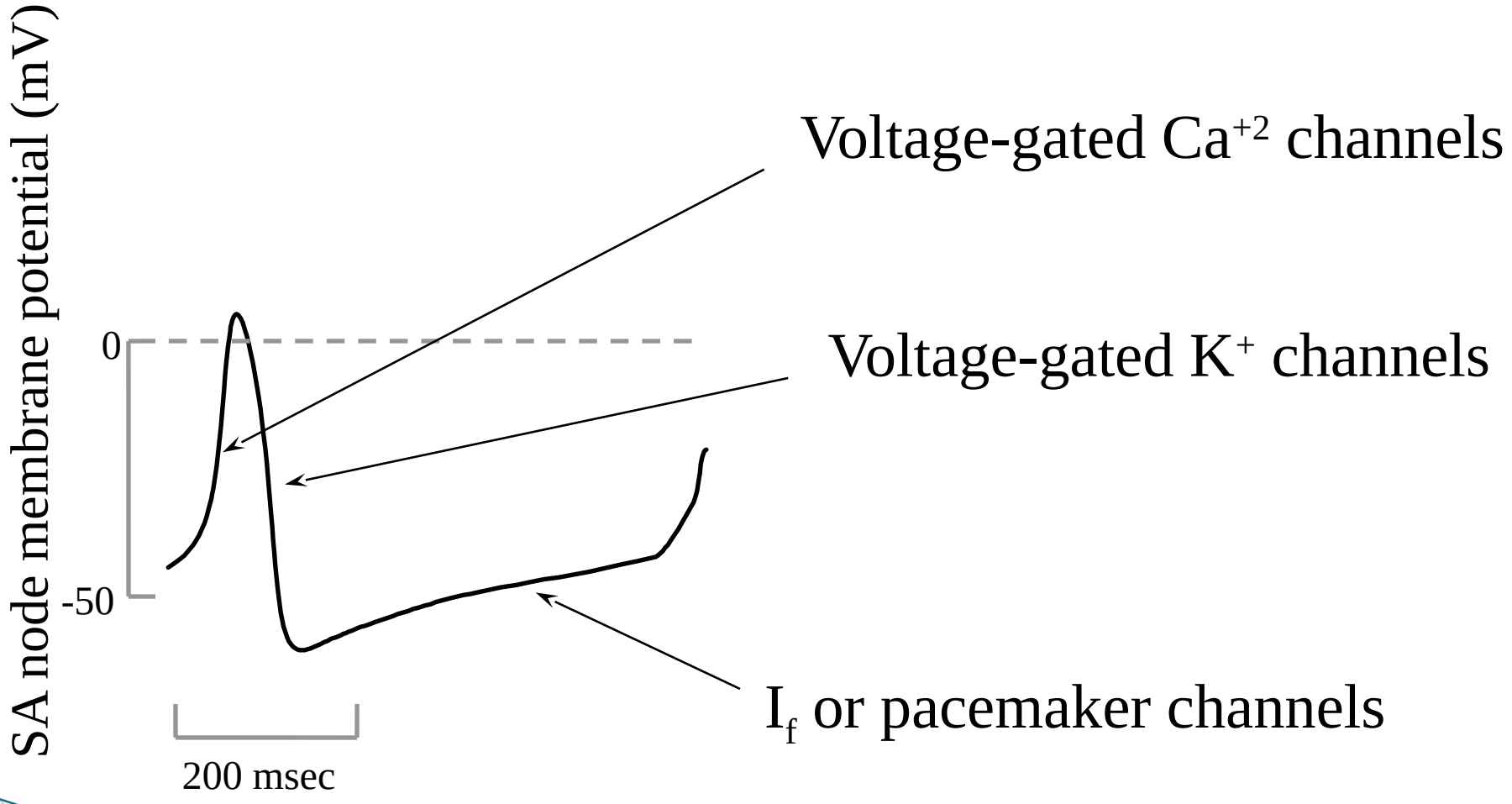
Exhibits:

1) The capture effect
2) Overdrive suppression

ACTION POTENTIAL OF CARDIAC MUSCLE CELLS (CONTRACTILE MUSCLES)

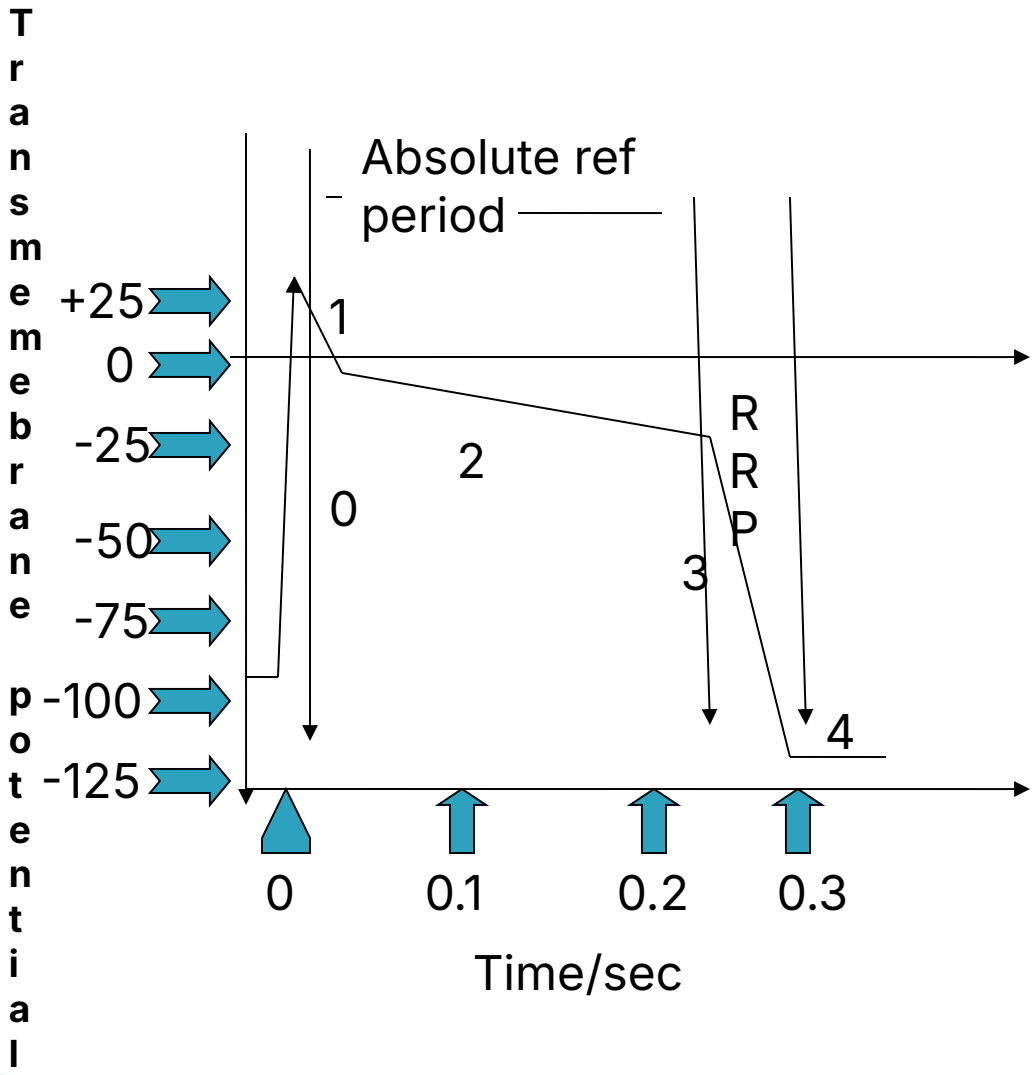


SA NODE ACTION POTENTIAL



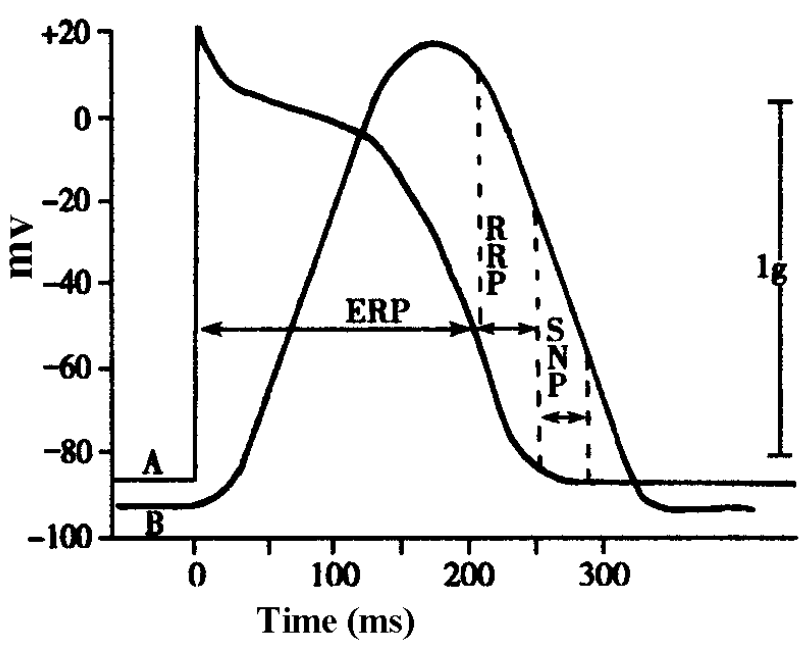
REFRACTORY PERIOD

- **Absolute Refractory Period** – regardless of the strength of a stimulus, the cell cannot be depolarized.
- **Relative Refractory Period** – stronger than normal stimulus can induce depolarization.



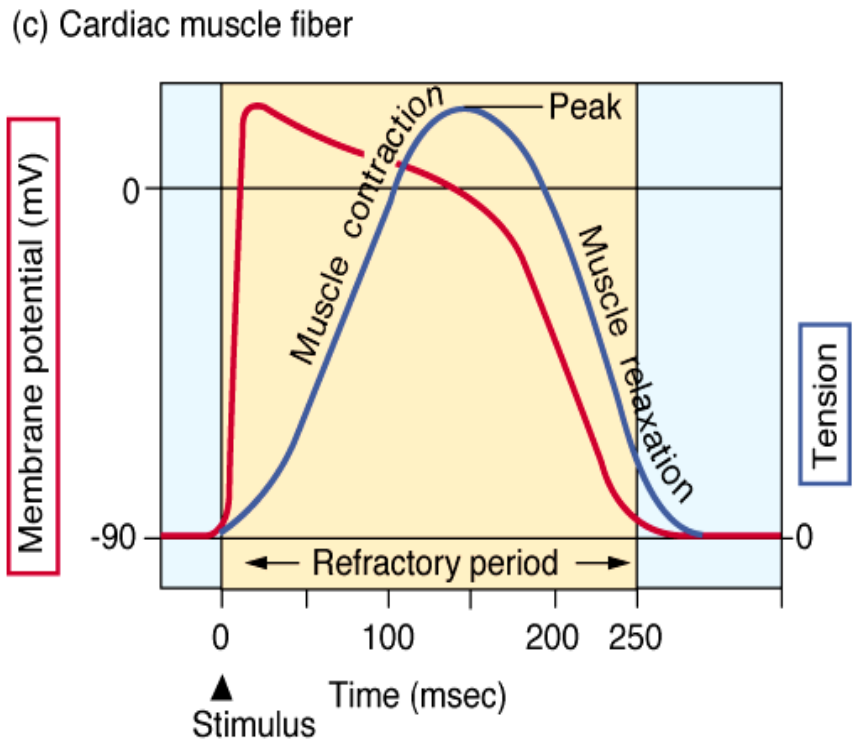
REFRACTORY PERIOD

- The plateau phase of the cardiac cell AP increases the duration of the AP to 300 msec,
- The refractory period of cardiac cells is long (250 msec).
  - compared to 1-5 msec in neurons and skeletal muscle fibers.



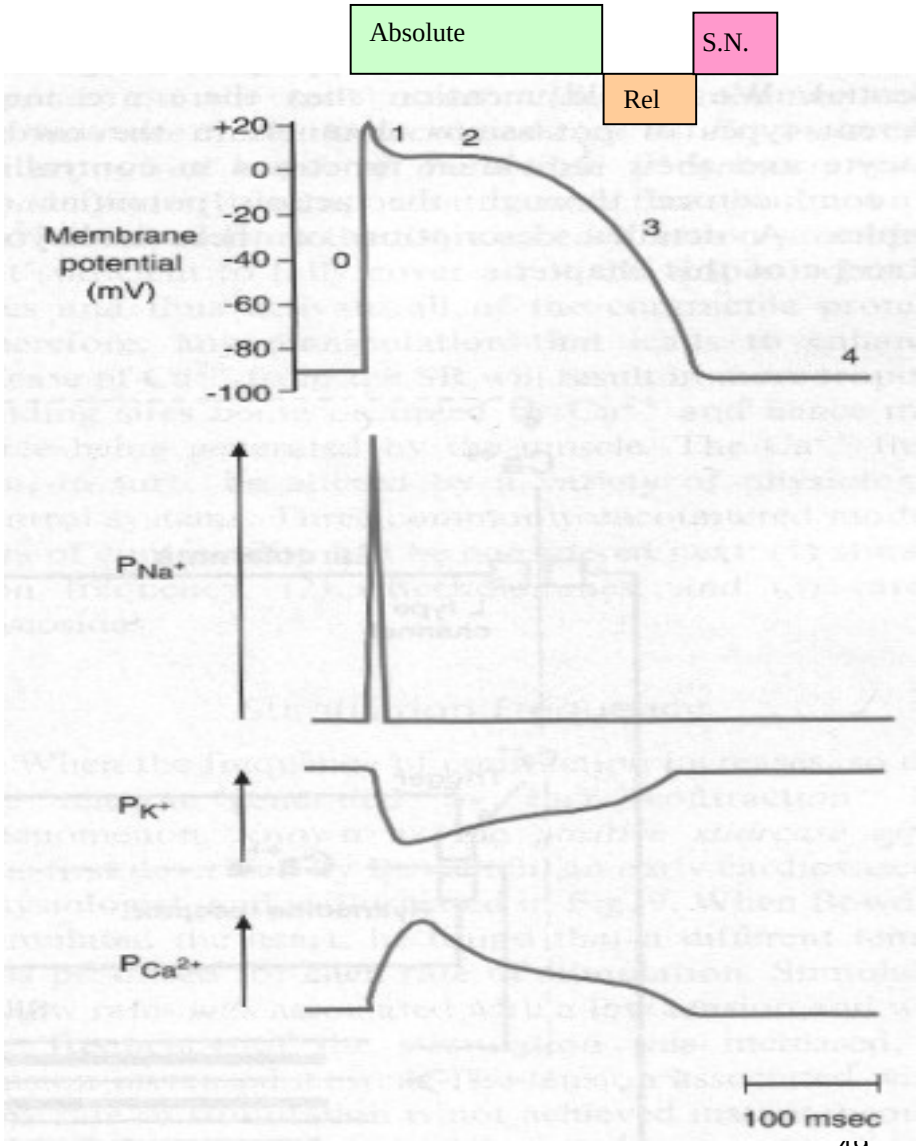
IMPORTANCE OF REFRACTORY PERIOD

- Long refractory period prevents tetanic contractions
- systole and diastole occur alternately.
- It is very important for pumping blood to arteries.

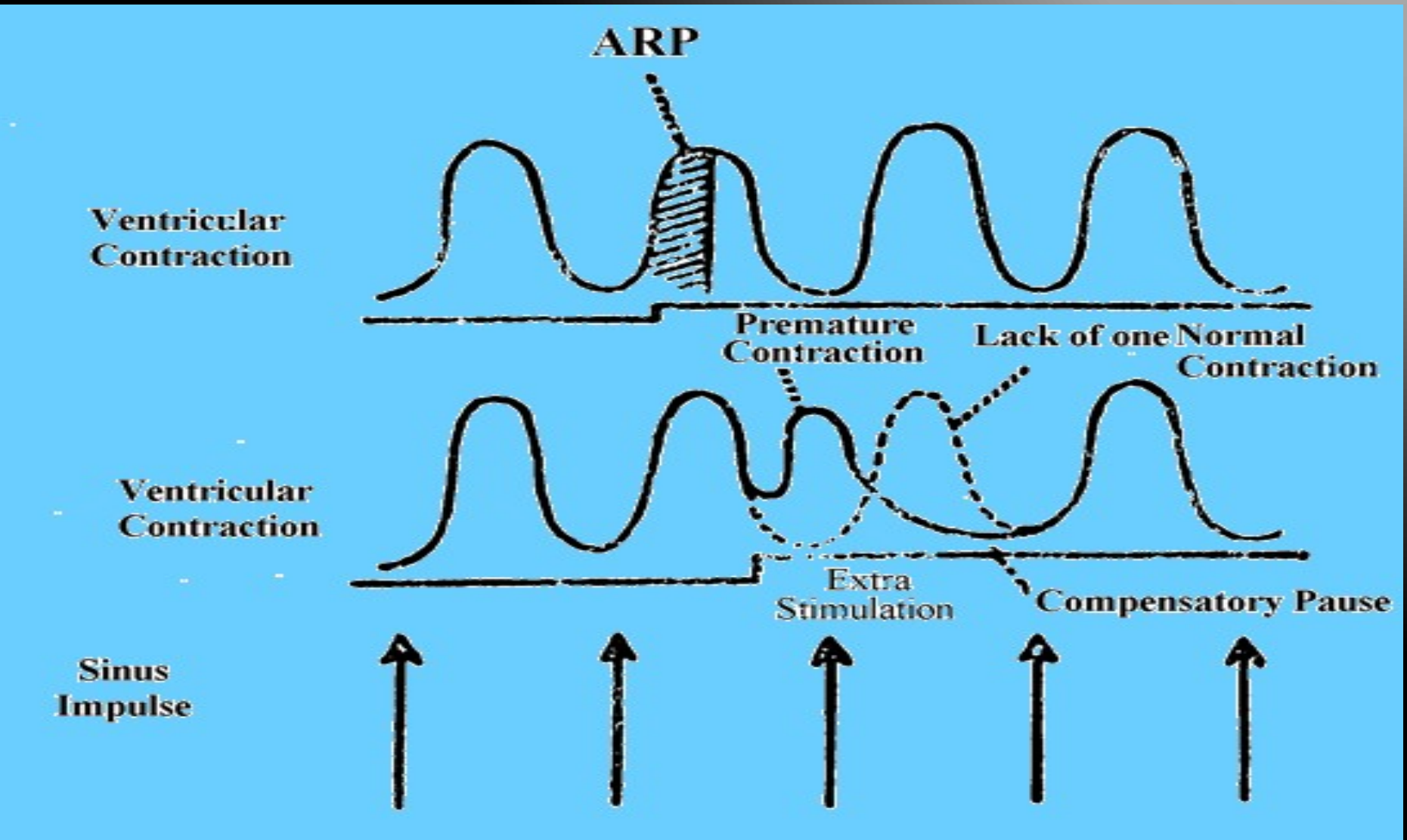


SUPRANORMAL PERIOD

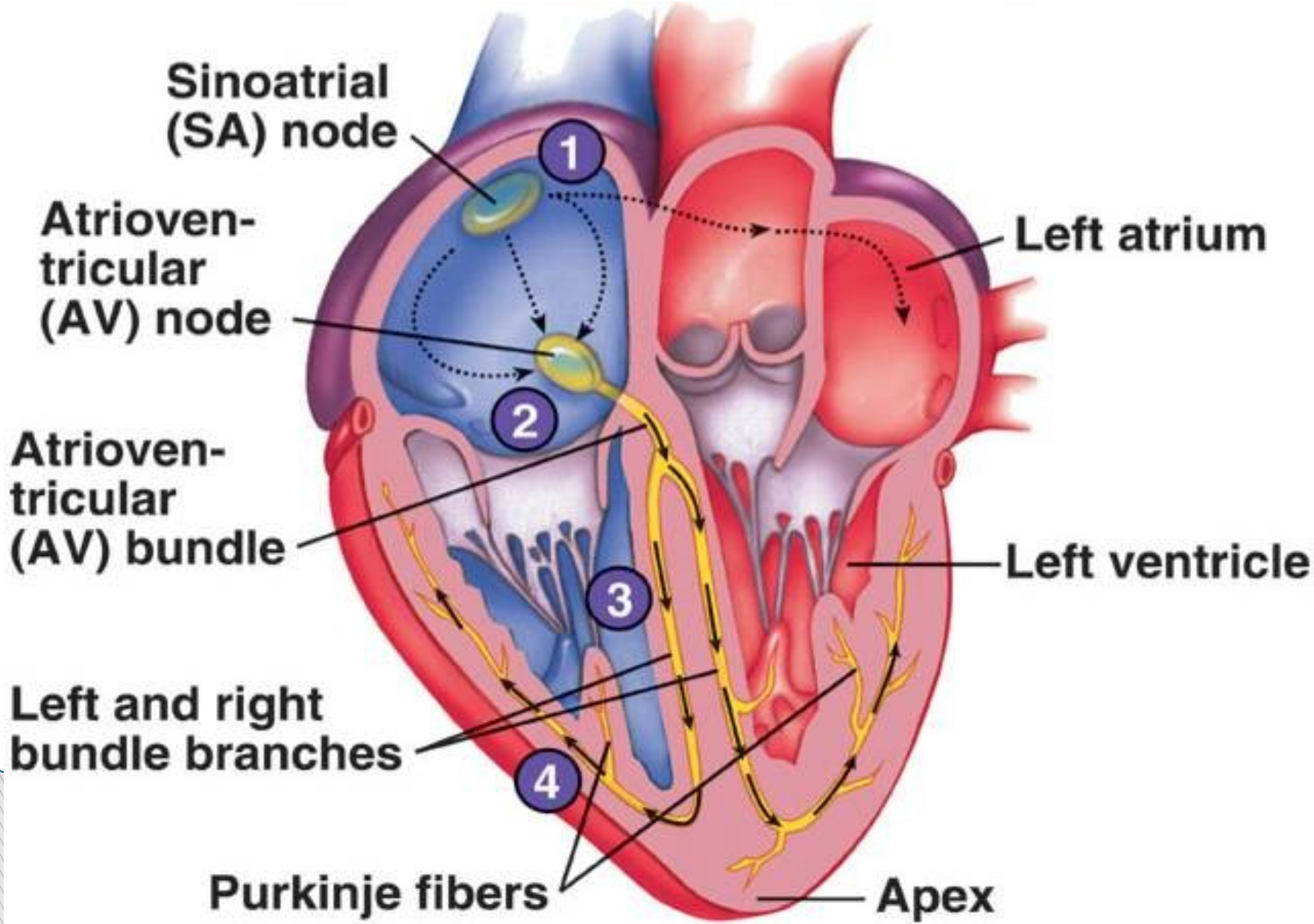
- **The cells can be restimulated and the threshold is actually lower than normal.**
- Usually accompanied by positive after-potentials as some potassium channels close.
- *Can be source of reentrant arrhythmias*

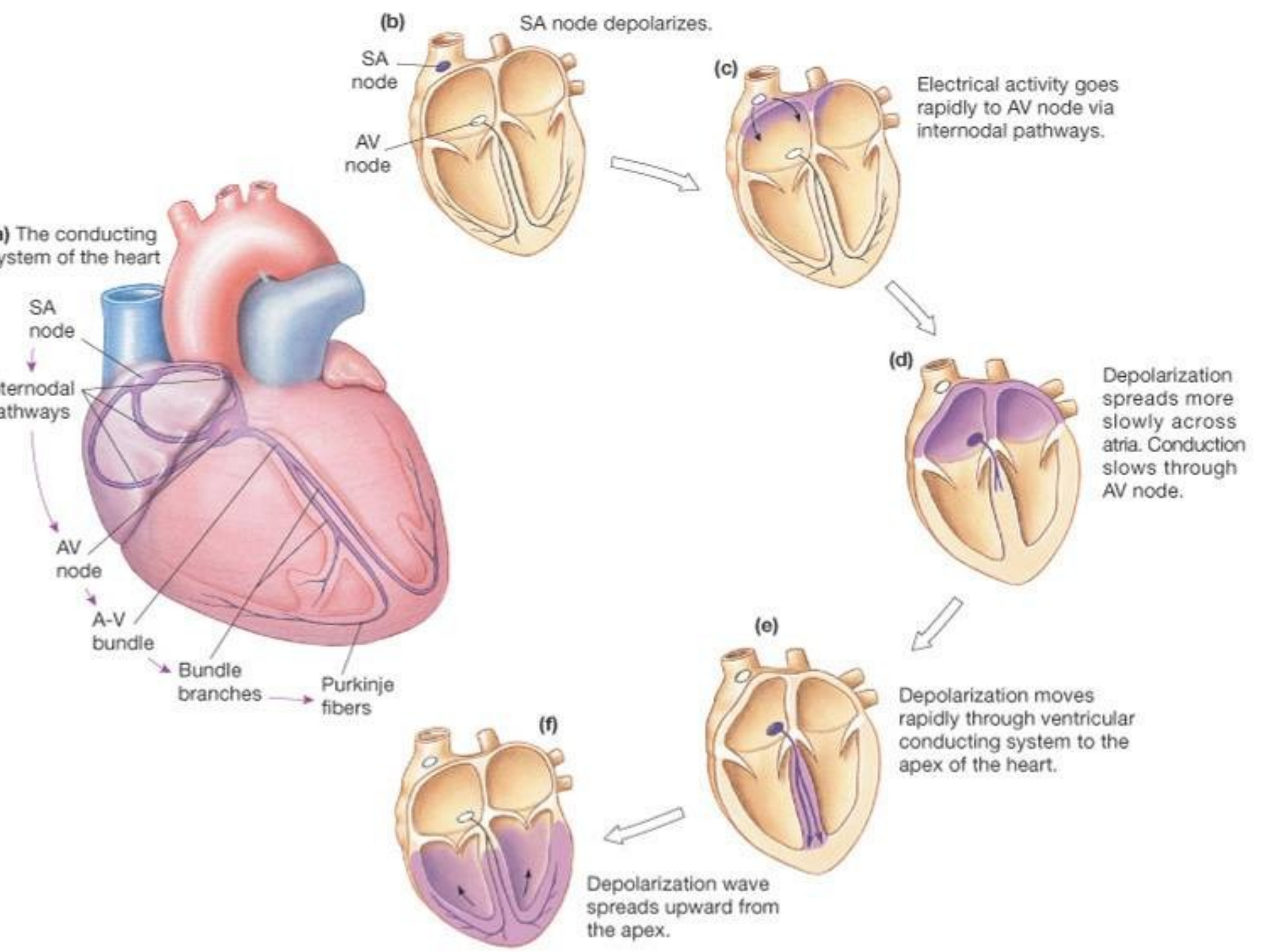


PREMATURE EXCITATION, PREMATURE CONTRACTION AND COMPENSATORY PAUSE



CONDUCTING SYSTEM OF HEART




(b)
SA node depolarizes.
SA node
AV node
(c)
Electrical activity goes rapidly to AV node via internodal pathways.
(d)
Depolarization spreads more slowly across atria. Conduction slows through AV node.
(e)
Depolarization moves rapidly through ventricular conducting system to the apex of the heart.
(f)
Depolarization wave spreads upward from the apex.
) The conducting ystem of the heart
SA node
ternodal athways
AV node
A-V bundle
Bundle branches
Purkinje fibers

FLOW OF CARDIAC ELECTRICAL ACTIVITY (ACTION POTENTIALS)

SA node	Pacing (sets heart rate)
↓	
Atrial Muscle	0.4m/s
↓	
AV node	0.02 m/s Delay
↓	
Purkinje System	4m/s Rapid, uniform spread
↓	
Ventricular Muscle	1m/s

CHARACTERISTICS OF CONDUCTION IN HEART

- **Delay in transmission at the A-V node** (150 –200 ms)
- This determines the sequence of the atrial and ventricular contraction

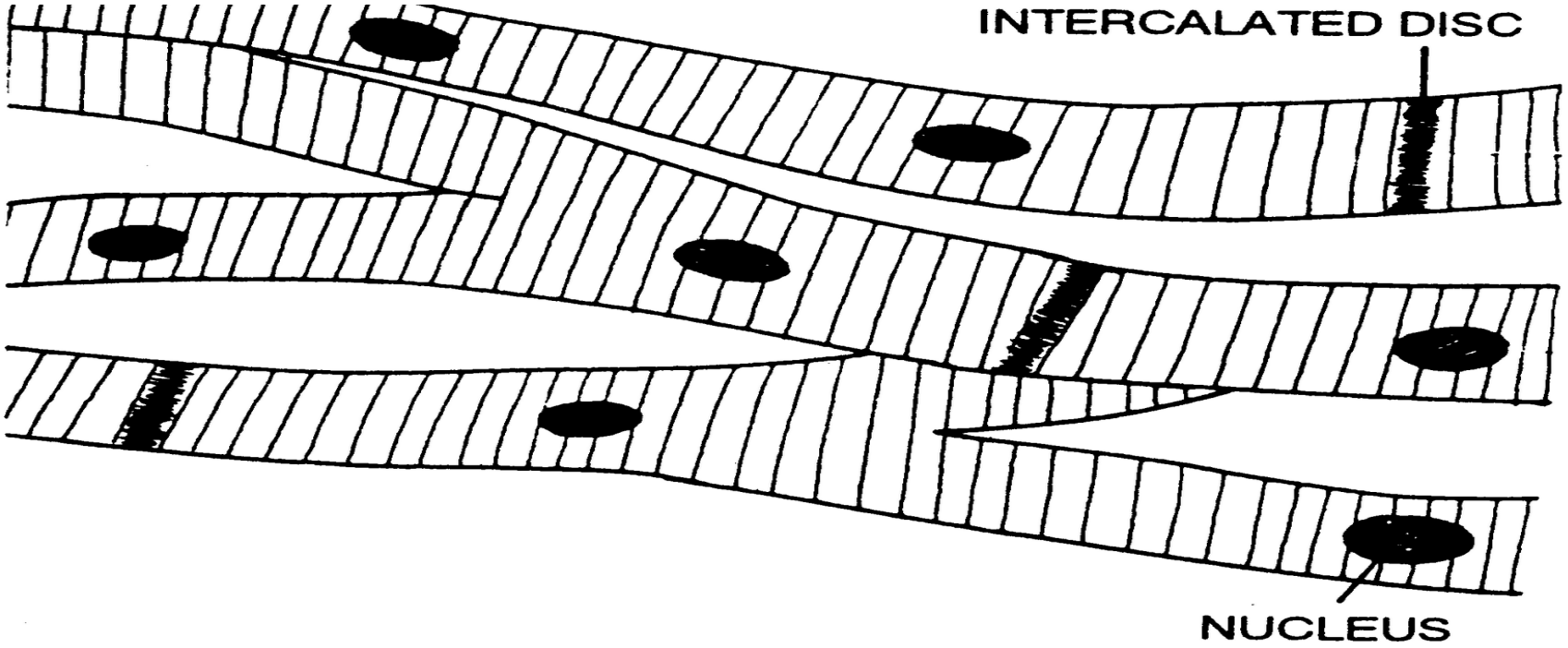
- **Rapid transmission of impulses in the Purkinje system**
- This synchronize contraction of entire ventricles

FACTORS DETERMINING CONDUCTIVITY

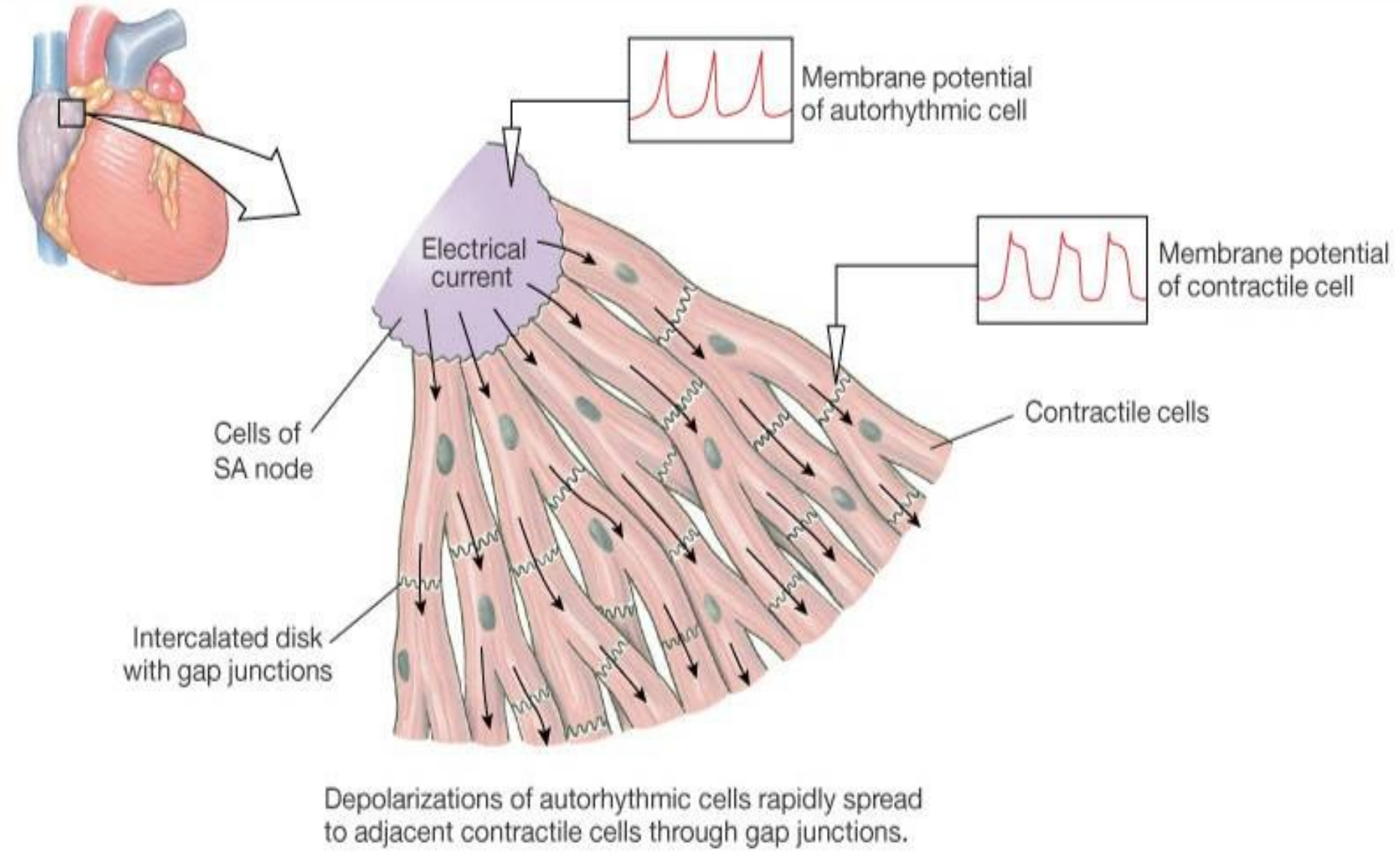
- Anatomical factors
- Physiological factors

ANATOMICAL FACTORS

- A.Gap junction between working cells
  - functional atrial and ventricular syncytium
- Diameter of the cardiac cell – causes conductive resistance –



COORDINATING THE PUMP: ELECTRICAL SIGNAL FLOW



Depolarizations of autorhythmic cells rapidly spread to adjacent contractile cells through gap junctions.

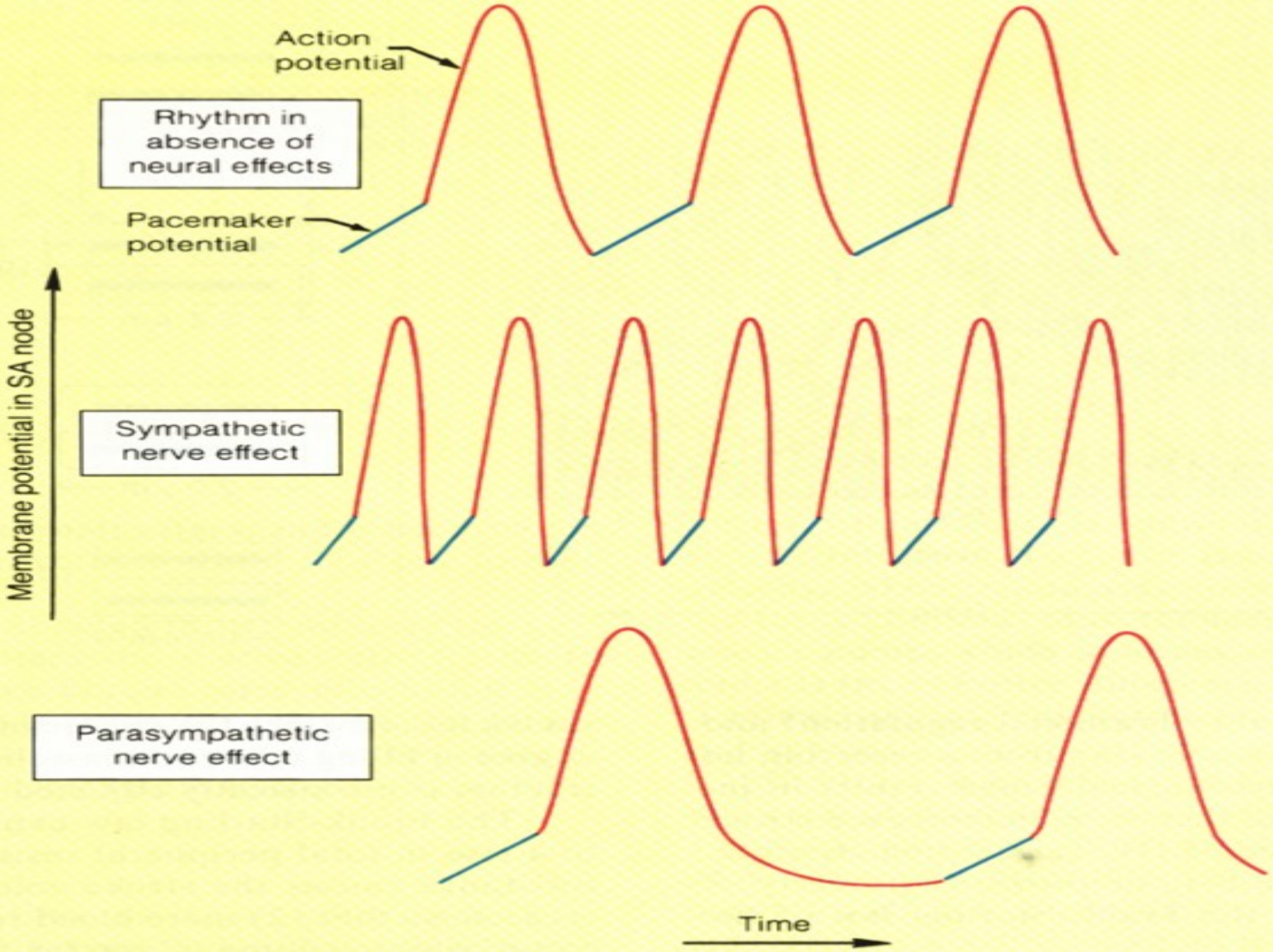
PHYSIOLOGICAL FACTORS

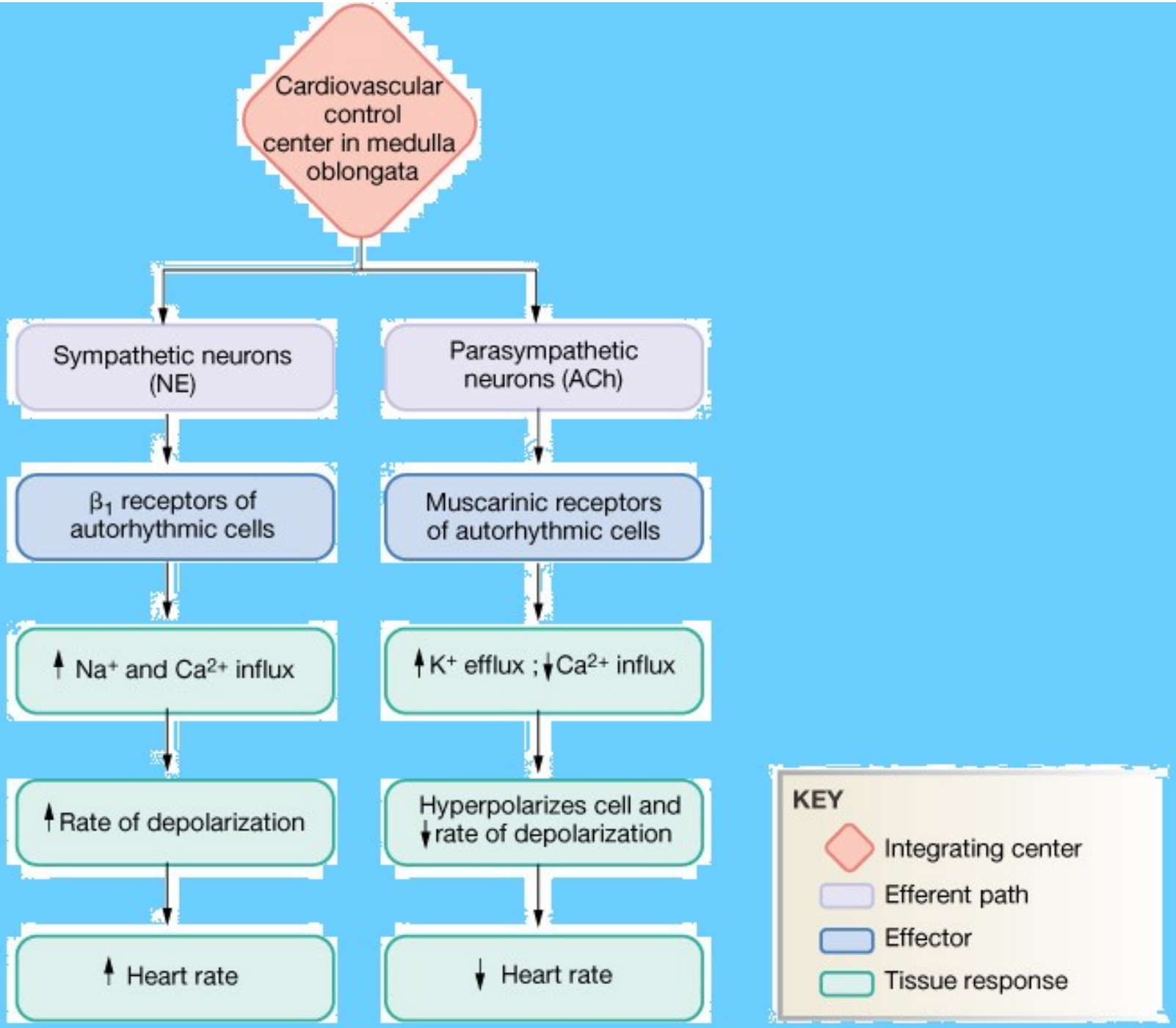
Neural and humoral control of the cardiac function

- Sympathetic
- parasympathetic
- T3, T4

EFFECT OF AUTONOMIC NERVE ACTIVITY ON THE HEART

Region Effected	Sympathetic Nerve	Parasympathetic Nerve
SA node	Increased rate of diastole depolarization ; increased cardiac rate	Decreased rate of diastole depolarization ; Decreased cardiac rate
AV node	Increase conduction rate	Decreased conduction rate
Atrial muscle	Increase strength of contraction	Decreased strength of contraction
Ventricular muscle	Increased strength of contraction	No significant effect


Action potential
Rhythm in absence of neural effects
Pacemaker potential
Membrane potential in SA node
Sympathetic nerve effect
Parasympathetic nerve effect
Time


Cardiovascular control center in medulla oblongata
Sympathetic neurons (NE)
Parasympathetic neurons (ACh)
β_1 receptors of autorhythmic cells
Muscarinic receptors of autorhythmic cells
↑ Na^+ and Ca^{2+} influx
↑ K^+ efflux ; ↓ Ca^{2+} influx
↑ Rate of depolarization
Hyperpolarizes cell and ↓ rate of depolarization
↑ Heart rate
↓ Heart rate
KEY
Integrating center
Efferent path
Effector
Tissue response

EFFECTS OF SYMPATHETIC NERVE

Postganglionic sympathetic nerves (epinephrine) and adrenal gland (epinephrine and norepinephrine)

Automaticity of SA node cell rise heart rate increase
→**Positive chronotropic effect**

Increase the conductivity of slow response cells

→**Positive dromotropic action**

Increase the Ca^{2+} concentration in plasma during excitation

→Increase myocardial contractility

→ **positive inotropic action**

Comparisons Among Skeletal, Smooth, and Cardiac Muscle

Property	Skeletal	Smooth (single-unit)	Smooth (multi-unit)	Cardiac
Striations (sarcomeres)	Yes	No	No	Yes
Actin and myosin	Yes	Yes	Yes	Yes
Level of control	Voluntary	Involuntary	Involuntary	Involuntary
Neural input	Somatic	Autonomic	Autonomic	Autonomic
Neuroeffector junction	Neuromuscular junction—specific	Varicosities—diffuse	Varicosities—diffuse	Varicosities—diffuse
Hormonal control	None	Several, depending on location	Several, depending on location	Epinephrine
Source of calcium	SR	SR and ECF	SR and ECF	SR and ECF
Regulatory protein that binds calcium	Troponin	Calmodulin	Calmodulin	Troponin
Gap junctions	No	Yes	No (or few)	Yes
Pacemaker activity	No	Yes	No	Yes
Myosin ATPase activity	Fastest	Slowest	Slowest	Intermediate
Recruitment	Yes	No	Yes	No

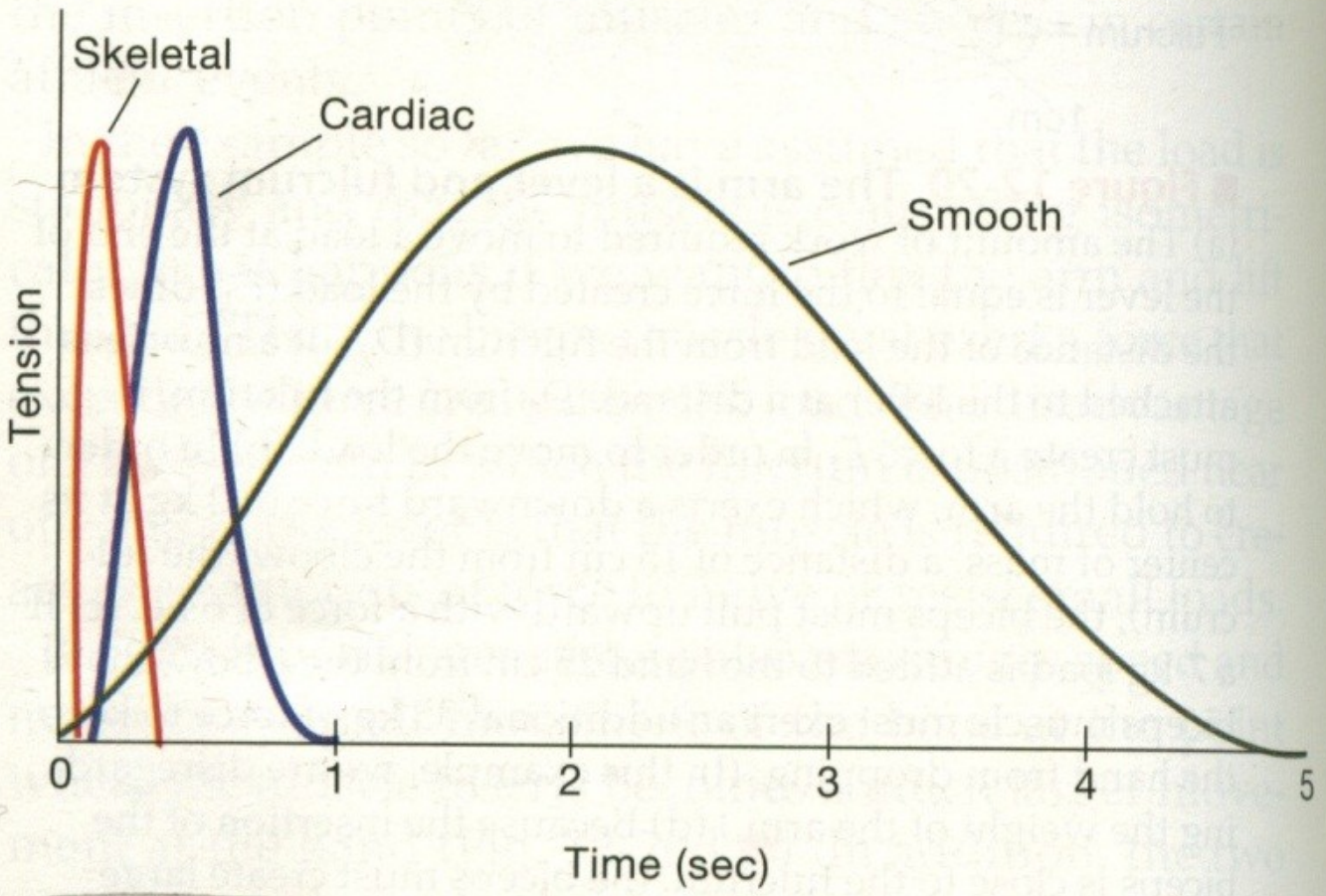


Figure 12-22 Duration of muscle contraction in three types of muscle Skeletal muscle has the quickest contraction and relaxation time. Smooth muscle is much slower to contract and relax. Tension development in a cardiac muscle twitch is more like that in a skeletal muscle twitch than in a smooth muscle twitch.

THANKYOU